

Experiment No. 12

Aim: To study the role of emulsifying agent in stabilizing the emulsion of an oil.

Apparatus: Beakers, test tube stand, test tubes, droppers, stopwatch, etc.

Chemicals: Soap solution detergent solution, paraffin oil, coconut oil, groundnut oil, linseed oil, engine oil, etc. (any two oils)

Theory:

An emulsion is a mixture of two immiscible liquids. Emulsions prepared by shaking of two liquids. They are unstable and tends to separate into two layers on standing. In order to get a stable emulsion, addition of substance is essential called emulsifying agent or emulsifier. Soap and detergent are acts as emulsifiers.

Soap and detergent are long chain compounds with polar groups. Emulsifier forms an interfacial film around the droplets of oil or water (disperse phase) and prevent the coalescence of droplets. Thus emulsion becomes stable.

Procedure:

Take 1 mL paraffin oil in a test tube and 1 mL groundnut oil in another test tube, then add 2 mL distilled water in each test tubes. Mark the liquid junction with marking pencil on both test tubes. Shake the mixture vigorously and put the test tubes in stands. Start the stopwatch and record the time required to separate two layers of water and oil in both the test tubes.

Add two drops of soap or detergent solution with the help of dropper in each test-tube containing mixture. Shake both test-tube vigorously and place in stand and start stop watch. Record the time required for complete separation of oil and water into two layers.

Observation Table:

Sr. No.	Emulsion	Time of separation (s)		
		I (t ₁)	II (t ₂)	Mean $\frac{(t_1+t_2)}{2}$
1	Paraffin oil + water	23	25	24
2	Paraffin oil + water + soap/ detergent	18	16	17
3	Groundnut oil + water	32	30	31
4	Groundnut oil+ water + soap/ detergent	17	16	15

Conclusion: More the time consumed for separation of emulsion more will be the stability of the emulsion.

Thus order of stability observed as follows.

Remark and sign of teacher:

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MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. Emulsion is colloidal system in which dispersed phase and dispersion medium are
 a. miscible liquids ✓ c. immiscible liquids
 b. miscible solids d. immiscible solid
2. Which of the following is **NOT** an emulsifier? ✓
 a. Soap b. Shampoo c. Detergent d. Charcoal
3. Pick out odd emulsifier.
 a. Milk b. Cod liver oil ✓ c. Oil paint d. Sodium lauryl sulfate
4. By addition of emulsifying agents emulsion becomes
 ✓ a. more stable c. partially stable
 b. unstable d. highly unstable
5. There are four emulsions namely A, B, C, D take different time to separates in their different phases. B takes more time than C, C takes more time than D, and D take more time than A. Which one is more stable emulsion among the given emulsions?
 a. A ✓ b. B c. C d. D
6. Detergents are used as best emulsifiers because they are
 a. non polar compound
 b. ionic inorganic compound
 ✓ c. covalent bonded organic compound
 d. long chain compound with polar groups.
7. Milk is
 ✓ a. an emulsion b. a gel c. a sol d. an aerosol
8. When small quantity of an oil paint is stirred in to water, the mixture will be a
 a. true solution c. suspension
 ✓ b. colloidal solution d. homogeneous solution

Short answer questions

1. What is an emulsion?

Ans. An emulsion is a colloidal system in which one liquid is dispersed in another immiscible liquid.

2. What is emulsifying agent?

Ans. An emulsifying agent is a substance that stabilizes an emulsion by preventing the dispersed liquid droplets from coalescing.

3. What are the types of emulsions?

Ans. The two types of emulsions are:
 Oil in Water (O/W) emulsion
 Water in Oil (W/O) emulsion

4. Give two examples of emulsifying agents.

Ans. Soap and detergent (e.g., sodium lauryl sulfate).

Remark and sign of teacher: